

UNIT-1

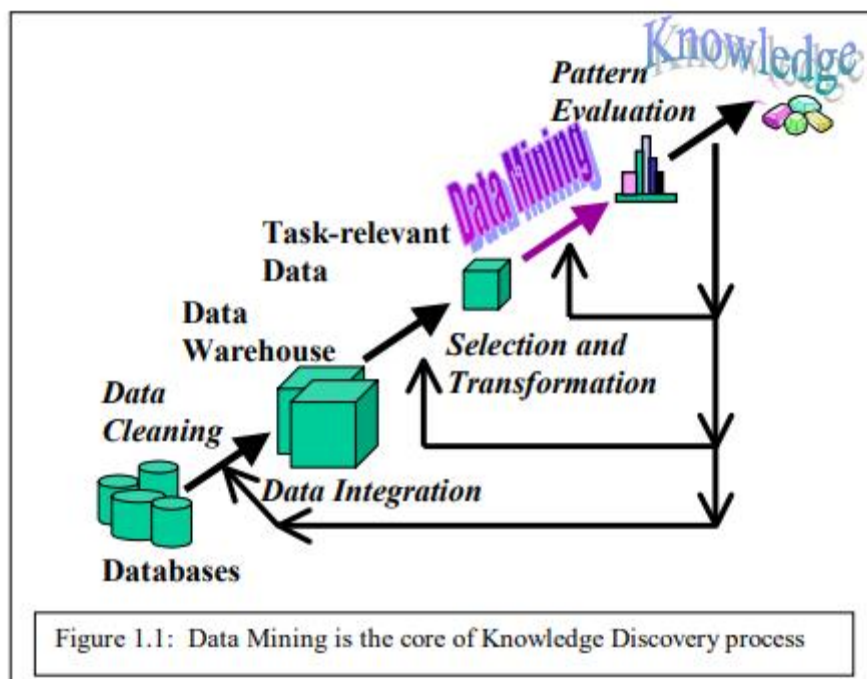
Data Mining Introduction:

Data mining-on what kind of data, Data mining functionalities, Technologies, Applications, Major issues in data mining, Data Objects and Attributes Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity.

What is data mining?

Data mining refers to extracting or mining" knowledge from huge amounts of data. There are many other terms related to data mining, such as knowledge mining, knowledge extraction, data/pattern analysis, data archaeology, and data dredging. Many people treat data mining as a synonym for another popularly used term, Knowledge Discovery in Databases", or KDD.

Essential step in the process of knowledge discovery in databases

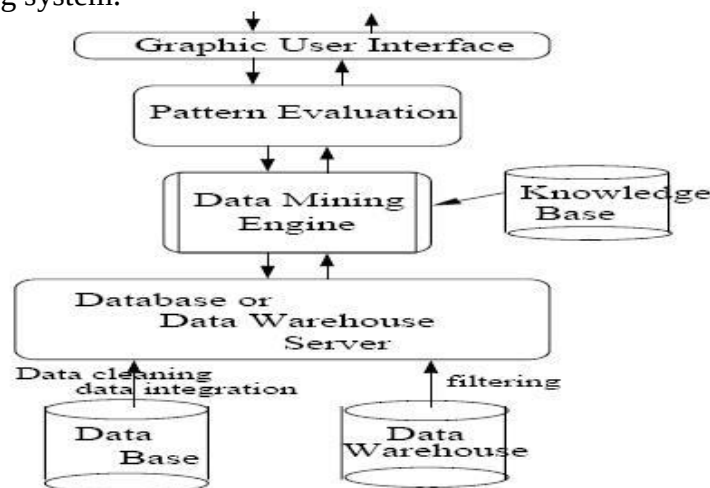


Knowledge discovery as a process is depicted in following figure and consists of an iterative sequence of the following steps.

- Data cleaning: to remove noise or irrelevant data.
- Data integration: where multiple data sources may be combined.
- Data selection: where data relevant to the analysis task are retrieved from the database.
- Data transformation: where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations.
- Data mining: an essential process where intelligent methods are applied in order to extract data patterns.
- Pattern evaluation to identify the truly interesting patterns representing knowledge based on some interestingness measures.
- Knowledge presentation: where visualization and knowledge representation techniques are used to present the mined knowledge to the user.

Architecture of a typical data mining system/Major Components

- Data mining is the process of discovering interesting knowledge from large amounts of data stored either in databases, data warehouses, or other information repositories. Based on this view, the architecture of a typical data mining system may have the following major components:
- A database, data warehouse, or other information repository, which consists of the set of databases, data warehouses, spreadsheets, or other kinds of information repositories containing the student and course information.
- A database or data warehouse server which fetches the relevant data based on users' data mining requests.
- A knowledge base that contains the domain knowledge used to guide the search or to evaluate the interestingness of resulting patterns. For example, the knowledge base may contain metadata which describes data from multiple heterogeneous sources.
- A data mining engine, which consists of a set of functional modules for tasks such as classification, association, classification, cluster analysis, and evolution and deviation analysis.
- A pattern evaluation module that works in tandem with the data mining modules by employing interestingness measures to help focus the search.
- A graphical user interface that allows the user an interactive approach to the data mining system.



Architecture of a typical data mining system.

What Kinds of Data Can Be Mined?

In general, data mining can be applied to any kind of data as long as the data are meaningful for a target application. The most basic forms of data for mining applications are

- database data
- data warehouse data and

- transactional data
- Data mining can also be applied to other forms of data
 - e.g., data streams, ordered/sequence data, graph or networked data, spatial data, text data, multimedia data.

1. Database Data

A database system, also called a **database management system (DBMS)**, consists of a collection of interrelated data, known as a **database**, and a set of software programs to manage and access the data. The software programs provide mechanisms for defining database structures and data storage; for specifying and managing concurrent, shared, or distributed data access; and for ensuring consistency and security of the information stored despite system crashes or attempts at unauthorized access.

A **relational database** is a collection of **tables**, each of which is assigned a unique name. Each table consists of a set of **attributes** (*columns* or *fields*) and usually stores a large set of **tuples** (*records* or *rows*). Each tuple in a relational table represents an object identified by a unique key and described by a set of attribute values. A semantic data model, such as an **entity-relationship (ER)** data model, is often constructed for relational databases. An ER data model represents the database as a set of entities and their relationships.

Example : A relational database for *AllElectronics*.

The company is described by the following relation tables:

customer, *item*, *employee*, and *branch*.

- The relation *customer* consists of a set of attributes describing the customer information, including a unique customer identity number (*cust ID*), customer name, address, age, occupation, annual income, credit information, and category. Similarly, each of the relations *item*, *employee*, and *branch* consists of a set of attributes describing the properties of these entities.
- Tables can also be used to represent the relationships between or among multiple entities. In our example, these include *purchases* (customer purchases items, creating a sales transaction handled by an employee), *items sold* (lists items sold in a given transaction), and *works at* (employee works at a branch of *AllElectronics*).

customer (*cust ID*, *name*, *address*, *age*, *occupation*, *annual income*, *credit information*, *category*, . . .)

item (*item ID*, *brand*, *category*, *type*, *price*, *place made*, *supplier*, *cost*, . . .)

employee (*empl ID*, *name*, *category*, *group*, *salary*, *commission*, . . .)

branch (*branch ID*, *name*, *address*, . . .)

purchases (*trans ID*, *cust ID*, *empl ID*, *date*, *time*, *method paid*, *amount*)

items sold (*trans ID*, *item ID*, *qty*)

works_at (*empl ID*, *branch ID*)

- Relational data can be accessed by **database queries** written in a relational query language (e.g., SQL) or with the assistance of graphical user interfaces.
- A given query is transformed into a set of relational operations, such as join, selection, and projection, and is then optimized for efficient processing.
- A query allows retrieval of specified subsets of the data. Suppose that your job is to analyze the *AllElectronics* data. Through the use of relational queries, you can ask things like, “*Show me a list of all items that were sold in the last quarter.*”
- Relational languages also use aggregate functions such as sum, avg (average), count, max (maximum), and min (minimum).

- Using aggregates allows you to ask: “*Show me the total sales of the last month, grouped by branch,*” or “*How many sales transactions occurred in the month of December?*” or “*Which salesperson had the highest sales?*”

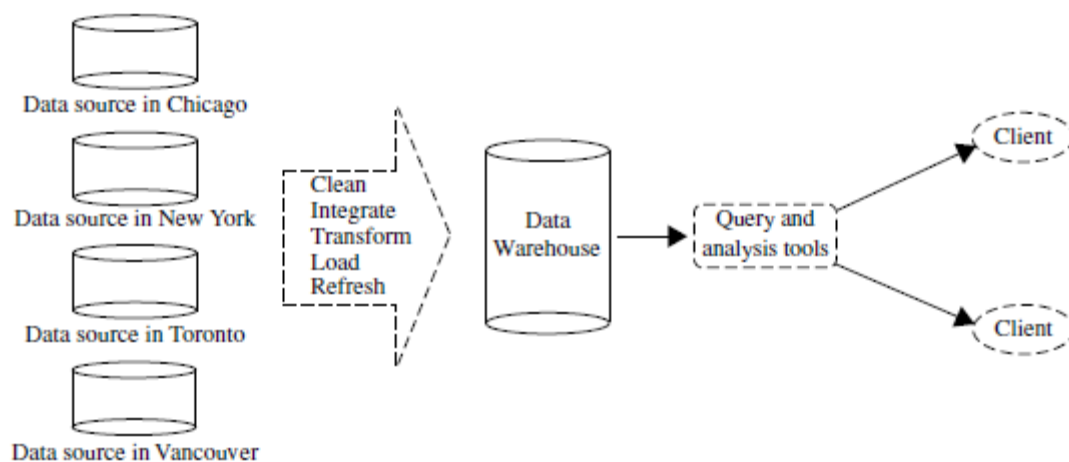
When **mining relational databases**,

- we can go further by **searching for trends or data patterns**.
 - For example, data mining systems can analyze customer data to predict the credit risk of new customers based on their income, age, and previous credit information.
- Data mining systems may also detect deviations—that is, items with sales that are far from those expected in comparison with the previous year. Such deviations can then be further investigated.
 - For example, data mining may discover that there has been a change in packaging of an item or a significant increase in price.
- Relational databases are one of the most commonly available and richest information repositories, and thus they are a major data form in the study of data mining.

DataWarehouses

A **data warehouse** is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site. Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.

Suppose that *AllElectronics* is a successful international company with branches around the world. Each branch has its own set of databases. The president of *AllElectronics* has asked you to provide an analysis of the company’s sales per item type per branch for the third quarter. This is a difficult task, particularly since the relevant data are spread out over several databases physically located at numerous sites. If *AllElectronics* had a data warehouse, this task would be easy.



Typical framework of a data warehouse for *AllElectronics*.

To facilitate decision making, the data in a data warehouse are organized around *major subjects* (e.g., customer, item, supplier, and activity). The data are stored to provide information from a *historical perspective*, such as in the past 6 to 12 months, and are typically **summarized**. For example, rather than storing the details of each sales transaction, the data warehouse may store a summary of the transactions per item type for each store or, summarized to a higher level, for each sales region.

A data warehouse is usually modeled by a multidimensional data structure, called a

data cube, in which each **dimension** corresponds to an attribute or a set of attributes in the schema, and each **cell** stores the value of some aggregate measure such as *count* or *sum.sales amount*. A data cube provides a multidimensional view of data and allows the pre computation and fast access of summarized data.

Example :A data cube for AllElectronics.

A data cube for summarized sales data of AllElectronics is presented in Figure.

The cube has three dimensions:

- *address* (with city values *Chicago, New York, Toronto, Vancouver*),
- *time* (with quarter values *Q1, Q2, Q3, Q4*), and
- *item*(with itemtype values *home entertainment, computer, phone, security*).

The aggregate value stored in each cell of the cube is *sales amount* (in thousands). For example, the total sales for the first quarter, *Q1*, for the items related to security systems in Vancouver is \$400,000, as stored in cell *hVancouver, Q1, security*. Additional cubes may be used to store aggregate sums over each dimension, corresponding to the aggregate values obtained using different SQL group-bys (e.g., the total sales amount per city and quarter, or per city and item, or per quarter and item, or per each individual dimension).

By providing multidimensional data views and the pre computation of summarized data, data warehouse systems can provide inherent support for OLAP. Online analytical processing operations make use of background knowledge regarding the domain of the data being studied to allow the presentation of data at *different levels of abstraction*. Such operations accommodate different user viewpoints.

Examples of OLAP operations include

- **drill-down** and
- **roll-up**,

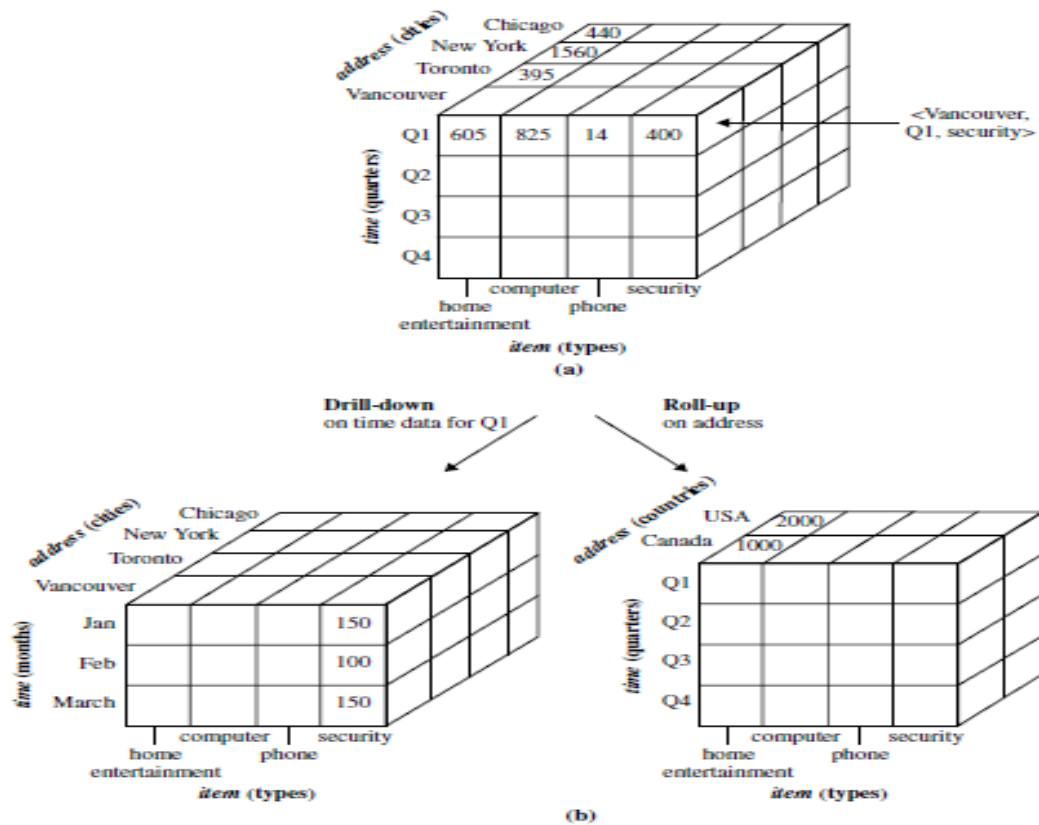
which allow the user to view the data at differing degrees of summarization, as illustrated in Figure 1.7(b).

- For instance, we can **drill down** on sales data summarized by *quarter* to see data summarized by *month*.
- Similarly, we can **roll up** on sales data summarized by *city* to view data summarized by *country*.

Although data warehouse tools help support data analysis, additional tools for data mining are often needed for in-depth analysis.

- **Multidimensional data mining** performs data mining in multidimensional space in an OLAP style. That is, it allows the exploration of multiple combinations of dimensions at varying levels of granularity in data mining, and thus has greater potential for discovering interesting patterns representing knowledge.

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A multidimensional data cube, commonly used for data warehousing, (a) showing summarized data for *AllElectronics* and (b) showing summarized data resulting from drill-down and roll-up operations on the cube in (a). For improved readability, only some of the cube cell values are shown.

Transactional Data

In general, each record in a **transactional database** captures a transaction, such as a customer's purchase, a flight booking, or a user's clicks on a web page. A transaction typically includes a unique transaction identity number (*trans ID*) and a list of the **items** making up the transaction, such as the items purchased in the transaction.

A transactional database may have additional tables, which contain other information related to the transactions, such as item description, information about the salesperson or the branch, and so on. The transactional database is usually either stored in a flat file in a format similar to the table.

A traditional database system is not able to perform market basket data analysis. Fortunately, data mining on transactional data can do so by mining *frequent itemsets*, that is, sets of items that are frequently sold together.

<i>trans_ID</i>	<i>list_of_item_IDs</i>
T100	11, 13, 18, 116
T200	12, 18
...	...

Data mining functionalities/Data mining tasks: what kinds of patterns can be mined?

Data mining functionalities are used to specify the kind of patterns to be found in data mining tasks. In general, data mining tasks can be classified into two categories:

- Descriptive
- predictive

Descriptive mining tasks characterize the general properties of the data in the database. Predictive mining tasks perform inference on the current data in order to make predictions.

Describe data mining functionalities, and the kinds of patterns they can discover
(or)

Define each of the following data mining functionalities: characterization, discrimination, association and correlation analysis, classification, prediction, clustering, and evolution analysis. Give examples of each data mining functionality, using a real-life database that you are familiar with.

1. Concept/class description: characterization and discrimination

Data can be associated with classes or concepts. It describes a given set of data in a concise and summarative manner, presenting interesting general properties of the data. These descriptions can be derived via data characterization, by summarizing the data of the class under study (often called the target class) data discrimination, by comparison of the target class with one or a set of comparative classes both data characterization and discrimination.

Data characterization

It is a summarization of the general characteristics or features of a target class of data.

Example:

A data mining system should be able to produce a description summarizing the characteristics of a student who has obtained more than 75% in every semester; the result could be a general profile of the student.

Data Discrimination is a comparison of the general features of target class data objects with the general features of objects from one or a set of contrasting classes.

Example:

The general features of students with high GPA's may be compared with the general features of students with low GPA's. The resulting description could be a general comparative profile of the students such as 75% of the students with high GPA's are fourth-year computing science students while 65% of the students with low GPA's are not.

The output of data characterization can be presented in various forms. Examples include pie charts, bar charts, curves, multidimensional data cubes, and multidimensional tables, including crosstabs. The resulting descriptions can also be presented as generalized relations, or in rule form called characteristic rules.

Discrimination descriptions expressed in rule form are referred to as discriminate rules.

2 Mining Frequent Patterns, Association and Correlations

It is the discovery of association rules showing attribute-value conditions that occur frequently together in a given set of data. For example, a data mining system may find association rules like

major(X, "computing science") $\Rightarrow$ owns(X, "personal computer")

[support = 12%, confidence = 98%]

Where, X is a variable representing a student. The rule indicates that of the students under study, 12% (support) major in computing science and own a personal computer. There is a 98% probability (confidence, or certainty) that a student in this group owns a personal computer.

Example:

A grocery store retailer to decide whether to put bread on sale. To help determine the impact of this decision, the retailer generates association rules that show what other products are frequently purchased with bread. He finds 60% of the times that bread is sold so are pretzels and that 70% of the time jelly is also sold. Based on these facts, he tries to capitalize on the association between bread, pretzels, and jelly by placing some pretzels and jelly at the end of the aisle where the bread is placed. In addition, he decides not to place either of these items on sale at the same time.

3 Classification and prediction

Classification:

- It predicts categorical class labels.
- It classifies data (constructs a model) based on the training set and the values (class labels) in a classifying attribute and uses it in classifying new data.
- Typical Applications
 - credit approval
 - target marketing
 - medical diagnosis
 - treatment effectiveness analysis

Classification can be defined as the process of finding a model (or function) that describes and distinguishes data classes or concepts, for the purpose of being able to use the model to predict the class of objects whose class label is unknown. The derived model is based on the analysis of a set of training data (i.e., data objects whose class label is known).

Example:

An airport security screening station is used to determine if passengers are potential terrorists or criminals. To do this, the face of each passenger is scanned and its basic pattern (distance between eyes, size, and shape of mouth, head etc) is identified. This pattern is compared to entries in a database to see if it matches any patterns that are associated with known offenders.

A classification model can be represented in various forms, such as

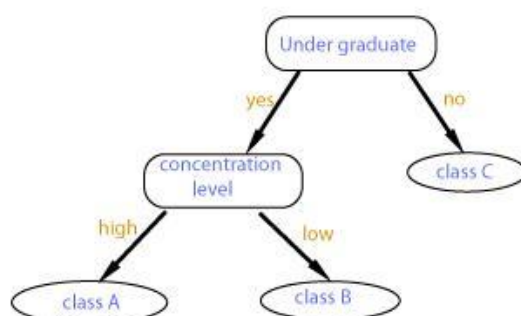
1) IF-THEN rules,

student (class, "undergraduate") AND concentration (level, "high") ==> class A

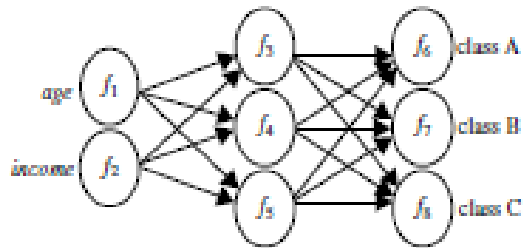
student (class, "undergraduate") AND concentration (level, "low") ==> class B

student (class, "post graduate") ==> class C

2) Decision tree



3) Neural Network



Prediction:

Find some missing or unavailable data values rather than class labels referred to as prediction. Although prediction may refer to both data value prediction and class label prediction, it is usually confined to data value prediction and thus is distinct from classification. Prediction also encompasses the identification of distribution trends based on the available data.

Example:

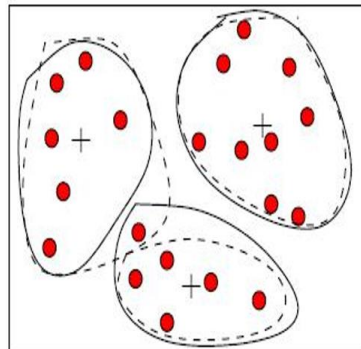
Predicting flooding is difficult problem. One approach is uses monitors placed at various points in the river. These monitors collect data relevant to flood prediction: water level, rain amount, time, humidity etc. These water levels at a potential flooding point in the river can be predicted based on the data collected by the sensors upriver from this point. The prediction must be made with respect to the time the data were collected.

Classification vs. Prediction

Classification differs from prediction in that the former is to construct a set of models (or functions) that describe and distinguish data class or concepts, whereas the latter is to predict some missing or unavailable, and often numerical, data values. Their similarity is that they are both tools for prediction: Classification is used for predicting the class label of data objects and prediction is typically used for predicting missing numerical data values.

4 Clustering analysis

- Clustering analyzes data objects without consulting a known class label.
- The objects are clustered or grouped based on the principle of maximizing the intraclass similarity and minimizing the interclass similarity.
- Each cluster that is formed can be viewed as a class of objects.
- Clustering can also facilitate taxonomy formation, that is, the organization of observations into a hierarchy of classes that group similar events together as shown below:



Example:

A certain national department store chain creates special catalogs targeted to various demographic groups based on attributes such as income, location and physical characteristics of potential customers (age, height, weight, etc). To determine the target mailings of the various catalogs and to assist in the creation of new, more specific catalogs, the company performs a clustering of potential customers based on the determined attribute values. The results of the clustering exercise are the used by management to create special catalogs and distribute them to the correct target population based on the cluster for that catalog.

Classification vs. Clustering

- In general, in classification you have a set of predefined classes and want to know which class a new object belongs to.
- Clustering tries to group a set of objects and find whether there is *some* relationship between the objects.
- In the context of machine learning, classification is *supervised learning* and clustering is *unsupervised learning*.

5 Outlier analysis:

A database may contain data objects that do not comply with general model of data. These data objects are outliers. In other words, the data objects which do not fall within the cluster will be called as outlier data objects. Noisy data or exceptional data are also called as outlier data. The analysis of outlier data is referred to as outlier mining.

Example

Outlier analysis may uncover fraudulent usage of credit cards by detecting purchases of extremely large amounts for a given account number in comparison to regular charges incurred by the same account. Outlier values may also be detected with respect to the location and type of purchase, or the purchase frequency.

6 Data evolution analysis describes and models regularities or trends for objects whose behavior changes over time.

Example:

The data of result the last several years of a college would give an idea if quality of graduated produced by it.

Correlation analysis

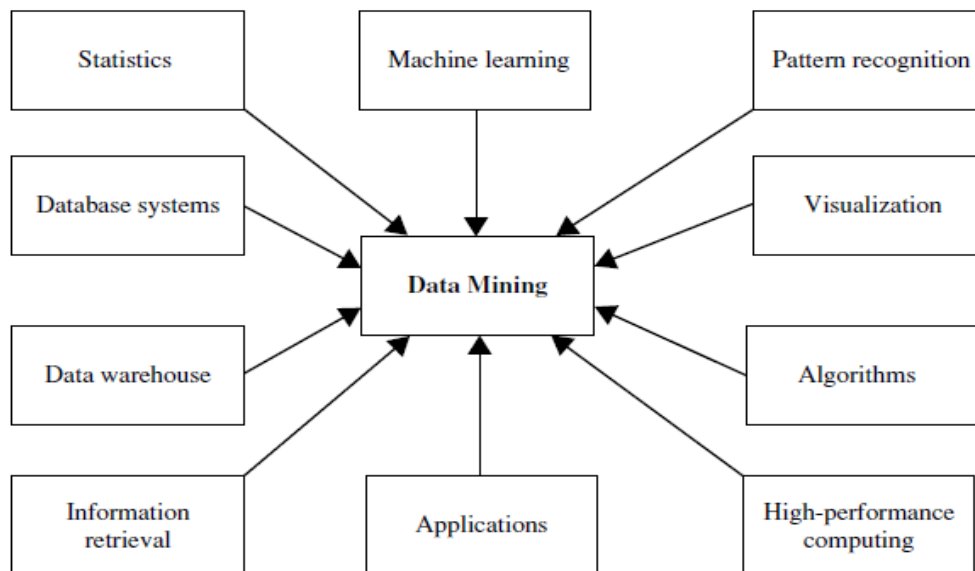
- Correlation analysis is a technique use to measure the association between two variables. A **correlation coefficient (r)** is a statistic used for measuring the strength of a supposed linear association between two variables. Correlations range from -1.0 to +1.0 in value.
 - A correlation coefficient of 1.0 indicates a perfect positive relationship in which high values of one variable are related perfectly to high values in the

other variable, and conversely, low values on one variable are perfectly related to low values on the other variable.

- A correlation coefficient of 0.0 indicates no relationship between the two variables. That is, one cannot use the scores on one variable to tell anything about the scores on the second variable.
- A correlation coefficient of -1.0 indicates a perfect negative relationship in which high values of one variable are related perfectly to low values in the other variables, and conversely, low values in one variable are perfectly related to high values on the other variable.

Which Technologies Are Used?

- Data mining has incorporated many techniques from other domains such as statistics, machine learning, pattern recognition, database and data warehouse systems, information retrieval, visualization, algorithms, high performance computing, and many application domains.
- The interdisciplinary nature of data mining research and development contributes significantly to the success of data mining and its extensive applications.



Data mining adopts techniques from many domains

Technologies Used in Data Mining

Data mining is a multidisciplinary field that integrates techniques from various domains. The major technologies involved include:

1. Database Technology

- Relational Databases: Store structured data in tables. Data mining often starts with querying relational databases using SQL.
- Data Warehousing: Integrates data from multiple sources into a central repository (data warehouse) for analysis.
- OLAP (Online Analytical Processing):
 - Enables multidimensional analysis of data.
 - Supports operations like roll-up, drill-down, slicing, and dicing.
- Data Cubes: Multidimensional structures that allow fast access to summarized data.

2. Statistics

- Fundamental for:
 - Data analysis
 - Pattern evaluation
 - Hypothesis testing
- Descriptive Statistics: Mean, median, mode, variance, etc., used to summarize data.
- Inferential Statistics: Used to draw conclusions about the population from samples.

3. Machine Learning

- Key component in discovering patterns and building predictive models.
- Supervised Learning: Classification, regression.
- Unsupervised Learning: Clustering, dimensionality reduction.
- Reinforcement Learning: Learning from interactions (less common in traditional data mining).

4. Information Retrieval

- Focuses on obtaining relevant information from large collections of data (especially text).
- Techniques such as:
 - Keyword matching
 - Document similarity
 - Ranking
- Used in web mining and text mining.

5. Pattern Recognition

- Identifies regularities or patterns in data.
- Applied in:
 - Image analysis
 - Speech recognition
 - Handwriting recognition

6. High-Performance Computing

- Needed to process large volumes of data efficiently.
- Technologies:
 - Parallel computing
 - Distributed systems (e.g., Hadoop, Spark)
 - Cloud computing

7. Artificial Intelligence (AI)

- Enables intelligent behavior such as learning, reasoning, and problem-solving.
- Neural networks, genetic algorithms, and expert systems are AI-based tools used in data mining.

8. Visualization

- Helps present mined knowledge effectively.
- Tools include:
 - Charts, plots, dashboards
 - Interactive visual analytics
- Important for:
 - Pattern understanding
 - Decision support

Algorithms

Data mining algorithms are used to discover patterns, trends, and useful information from large datasets.

1. Classification Algorithms

Used to predict categorical labels.

Algorithm	Description	Example
Decision Trees (ID3, C4.5)	Tree-like model of decisions	Loan approval
Naive Bayes	Probabilistic classifier based on Bayes' Theorem	Spam detection
k-Nearest Neighbors (k-NN)	Classifies based on closest training samples	Handwriting recognition
Support Vector Machine (SVM)	Finds optimal hyperplane for classification	Face detection
Neural Networks (ANN)	Mimics human brain for complex pattern recognition	Image & speech recognition

2. Clustering Algorithms

Used to group similar data points (unsupervised learning).

Algorithm	Description	Example
k-Means	Partitions data into k clusters	Customer segmentation
DBSCAN	Density-based clustering	Spatial data analysis
Hierarchical Clustering	Builds nested clusters in tree form	Gene expression data

3. Association Rule Learning

Used to discover interesting relationships between variables.

Algorithm	Description	Example
Apriori	Finds frequent itemsets using support/confidence	Market basket analysis
FP-Growth	More efficient, uses a compact FP-tree	Product bundling

4. Regression Algorithms

Predict continuous values.

Algorithm	Description	Example
Linear Regression	Models linear relationship between input and output	Predicting house prices
Logistic Regression	For binary classification problems	Disease diagnosis

5. Anomaly Detection

Identifies rare data patterns (outliers).

Algorithm	Description	Example
Statistical Models	Assumes normal distribution	Fraud detection
Isolation Forest	Randomly isolates observations	Intrusion detection
One-Class SVM	Learns boundary for normal data	Network security

6. Dimensionality Reduction

Simplifies data by reducing number of features.

Algorithm	Description	Example
PCA (Principal Component Analysis)	Projects data to lower dimensions	Visualization, Noise removal
t-SNE	Nonlinear, preserves local structure	High-dimensional data visualization

Applications of Data Mining

Data mining is applied across a wide range of real-world domains:

1. Business Intelligence

- **Market Basket Analysis:** Discover product purchase patterns.
- **Customer Segmentation:** Grouping customers for targeted marketing.
- **Churn Prediction:** Predicting which customers are likely to leave.
- **Sales Forecasting:** Predicting future sales trends.

2. Healthcare

- Disease diagnosis and prognosis.
- Drug discovery and genomic data analysis.
- Patient risk prediction.

3. Finance

- Fraud detection (credit card, insurance).
- Risk assessment for loans.
- Stock market analysis.

4. Telecommunications

- Call pattern analysis.
- Fault diagnosis.
- Customer churn modeling.

5. E-Commerce & Web

- Recommender systems (Amazon, Netflix).
- Personalized advertising.
- Clickstream analysis.

6. Scientific Research

- Astronomy, climate modeling, physics experiments.
- Pattern recognition in large datasets (e.g., particle collisions).

7. Education

- Student performance prediction.
- Dropout risk identification.
- Adaptive learning systems.

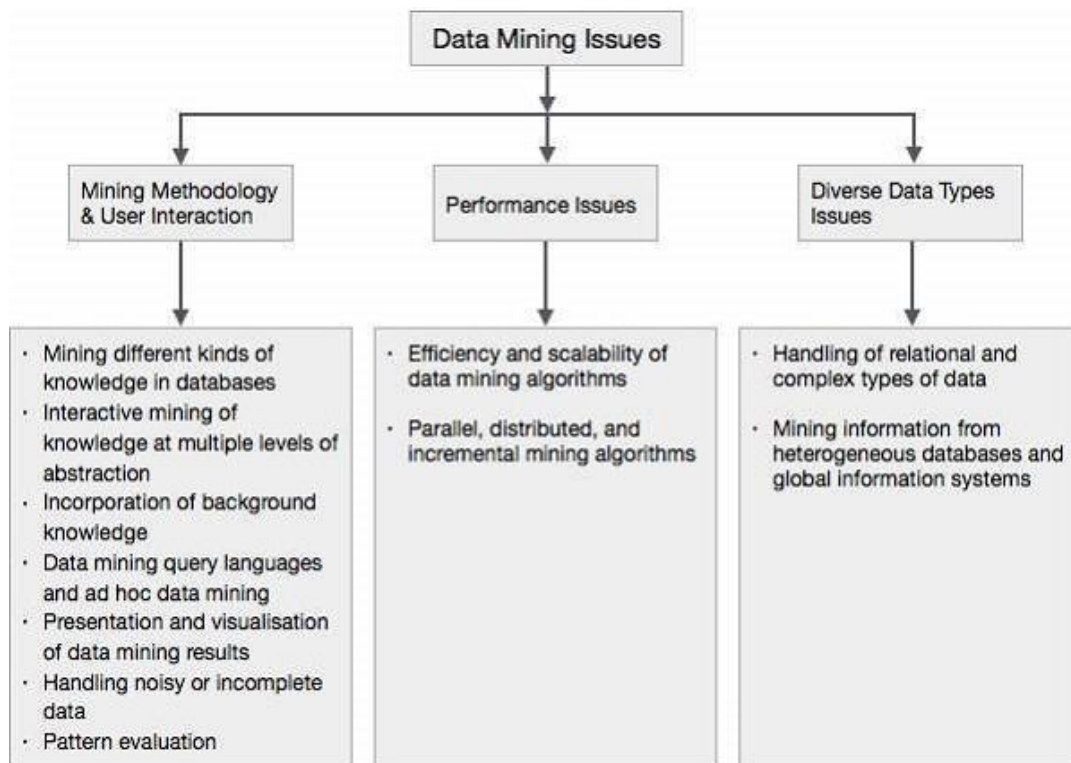
DATA MINING - ISSUES

Data mining is not an easy task, as the algorithms used can get very complex and data is not always available at one place. It needs to be integrated from various heterogeneous data sources. These factors also create some issues.

The major issues are

- Mining Methodology and User Interaction
- Performance Issues
- Diverse Data Types Issues

The following diagram describes the major issues.



Mining Methodology and User Interaction Issues

It refers to the following kinds of issues –

- **Mining different kinds of knowledge in databases** – Different users may be interested in different kinds of knowledge. Therefore, it is necessary for data mining to cover a broad range of knowledge discovery task.
- **Interactive mining of knowledge at multiple levels of abstraction** – The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on the returned results.
- **Incorporation of background knowledge** – To guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple levels of abstraction.
- **Data mining query languages and ad hoc data mining** – Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.
- **Presentation and visualization of data mining results** – Once the patterns are discovered it needs to be expressed in high level languages, and visual representations. These representations should be easily understandable.
- **Handling noisy or incomplete data** – The data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data cleaning methods are not there then the accuracy of the discovered patterns will be poor.
- **Pattern evaluation** – The patterns discovered should be interesting because either they represent common knowledge or lack novelty.

Performance Issues

There can be performance-related issues such as follows –

- **Efficiency and scalability of data mining algorithms** – In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.

- **Parallel, distributed, and incremental mining algorithms** – The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data into partitions which is further processed in a parallel fashion. Then the results from the partitions is merged. The incremental algorithms, update databases without mining the data again from scratch.

Diverse Data Types Issues

- **Handling of relational and complex types of data** – The database may contain complex data objects, multimedia data objects, spatial data, temporal data etc. It is not possible for one system to mine all these kind of data.
- **Mining information from heterogeneous databases and global information systems** – The data is available at different data sources on LAN or WAN. These data source may be structured, semi structured or unstructured. Therefore mining the knowledge from them adds challenges to data mining.

Data Objects and Attribute Types

Data objects

- Data sets are made up of data objects.
- A data object represents an entity.
- Examples:
 - sales database: customers, store items, sales
 - medical database: patients, treatments
 - university database: students, professors, courses
- Also called *samples*, *examples*, *instances*, *data points*, *objects*, *tuples*.
- Data objects are described by **attributes**.
- Database rows -> data objects; columns -> attributes.

Attributes

- Attribute (or dimensions, features, variables): a data field, representing a characteristic or feature of a data object.
 - *E.g., customer _ID, name, address*
- Attribute Types:
 - Nominal
 - Binary
 - Numeric: quantitative
 - Interval-scaled
 - Ratio-scaled
- **Nominal:** categories, states, or “names of things”
 - *Hair_color = {auburn, black, blond, brown, grey, red, white}*
 - marital status, occupation, ID numbers, zip codes
- **Binary**
 - Nominal attribute with only 2 states (0 and 1)
 - Symmetric binary: both outcomes equally important
 - e.g., gender
 - Asymmetric binary: outcomes not equally important.
 - e.g., medical test (positive vs. negative)
 - Convention: assign 1 to most important outcome (e.g., HIV positive)
- **Ordinal**
 - Values have a meaningful order (ranking) but magnitude between successive values is not known.
 - *Size = {small, medium, large}*, grades, army rankings
- **Numeric Attribute Types**

- Quantity (integer or real-valued)
- **Interval**
 - Measured on a scale of **equal-sized units**
 - Values have order
 - E.g., *temperature in C° or F°, calendar dates*
 - No true zero-point
- **Ratio**
 - Inherent **zero-point**
 - We can speak of values as being an order of magnitude larger than the unit of measurement (10 K° is twice as high as 5 K°).
 - e.g., *temperature in Kelvin, length, counts, monetary quantities*

Discrete vs. Continuous Attributes

- **Discrete Attribute**
 - Has only a finite or countably infinite set of values
 - E.g., zip codes, profession, or the set of words in a collection of documents
 - Sometimes, represented as integer variables
 - Note: Binary attributes are a special case of discrete attributes
- **Continuous Attribute**
 - Has real numbers as attribute values
 - E.g., temperature, height, or weight
 - Practically, real values can only be measured and represented using a finite number of digits
 - Continuous attributes are typically represented as floating-point variables

Measuring Data Similarity and Dissimilarity

Similarity and Dissimilarity

Distance or similarity measures are essential to solve many pattern recognition problems such as classification and clustering. Various distance/similarity measures are available in literature to compare two data distributions. As the names suggest, a similarity measures how close two distributions are. For multivariate data complex summary methods are developed to answer this question.

Similarity Measure

- Numerical measure of how alike two data objects are.
 - Value is higher when objects are more alike
 - Often falls between 0 (no similarity) and 1 (complete similarity).

Dissimilarity Measure

- Numerical measure of how different two data objects are.
 - Range from 0 (objects are alike) to ∞ (objects are different).
 - Lower when objects are more alike
 - Minimum dissimilarity is often 0
 - Upper limit varies

Proximity refers to a similarity or dissimilarity.

Data Matrix and Dissimilarity Matrix

- Data matrix
 - n data points with p dimensions
 - Two modes

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

■ Dissimilarity matrix

- n data points, but registers only the distance
- A triangular matrix
- Single mode

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

Similarity/Dissimilarity for Simple Attributes

Here, p and q are the attribute values for two data objects.

Attribute Type	Similarity	Dissimilarity
Nominal	$s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$	$d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$
Ordinal	$s = 1 - \frac{\ p-q\ }{n-1}$ (values mapped to integer 0 to n-1, where n is the number of values)	$d = \frac{\ p-q\ }{n-1}$
Interval or Ratio	$s = 1 - \ p-q\ , s = \frac{1}{1+\ p-q\ }$	$d = \ p-q\ $

1. Proximity Measure for Nominal Attributes

- Can take 2 or more states, e.g., red, yellow, blue, green (generalization of a binary attribute)
- Method 1: Simple matching
 - m : No. of matches,
 - p : total No. of variables

$$d(i, j) = \frac{p - m}{p}$$

- Method 2: Use a large number of binary attributes
 - creating a new binary attribute for each of the M nominal states

2. Proximity Measure for Binary Attributes

- A contingency table for binary data
- Distance measure for symmetric binary variables:
- Distance measure for asymmetric binary variables:
- Jaccard coefficient (*similarity* measure for *asymmetric* binary variables):

		Object <i>j</i>		
		1	0	sum
Object <i>i</i>	1	<i>q</i>	<i>r</i>	<i>q + r</i>
	0	<i>s</i>	<i>t</i>	<i>s + t</i>
sum		<i>q + s</i>	<i>r + t</i>	<i>p</i>

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

$$d(i, j) = \frac{r + s}{q + r + s}$$

$$sim_{Jaccard}(i, j) = \frac{q}{q + r + s}$$

$$coherence(i, j) = \frac{sup(i, j)}{sup(i) + sup(j) - sup(i, j)} = \frac{q}{(q + r) + (q + s) - q}$$

- Note: Jaccard coefficient is the same as “coherence”:

Dissimilarity between Binary Variables

Example:

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

- Gender is a symmetric attribute
- The remaining attributes are asymmetric binary
- Let the values Y and P be 1, and the value N 0

$$d(jack, mary) = \frac{0+1}{2+0+1} = 0.33$$

$$d(jack, jim) = \frac{1+1}{1+1+1} = 0.67$$

$$d(jim, mary) = \frac{1+2}{1+1+2} = 0.75$$

Dissimilarity of Numeric Data

- Distance measures that are commonly used for computing the dissimilarity of objects described by numeric attributes. These measures include the *Euclidean*, *Manhattan*, and *Minkowski distances*.
- In some cases, the data are normalized before applying distance calculations. This involves transforming the data to fall within a smaller or common range, such as [-1, 1] or [0.0, 1.0].
- Consider a *height* attribute, for example, which could be measured in either meters or inches. In general, expressing an attribute in smaller units will lead to a larger range for that attribute, and thus tend to give such attributes greater effect or “weight.”

- Normalizing the data attempts to give all attributes an equal weight. It may or may not be useful in a particular application

Common Properties of Dissimilarity Measures

- **Distance**, such as the Euclidean distance, is a dissimilarity measure and has some well known properties:
 - $d(p, q) \geq 0$ for all p and q , and $d(p, q) = 0$ if and only if $p = q$,
 - $d(p, q) = d(q, p)$ for all p and q ,
 - $d(p, r) \leq d(p, q) + d(q, r)$ for all p, q , and r , where $d(p, q)$ is the distance (dissimilarity) between points (data objects), p and q .
- A distance that satisfies these properties are called a **metric**. Following is a list of several common distance measures to compare multivariate data. We will assume that the attributes are all continuous.

Minkowski distance: A popular distance measure

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \dots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and h is the order (the distance so defined is also called L - h norm)

Special Cases of Minkowski Distance

- $h = 1$: Manhattan (city block, L_1 norm) distance
 - E.g., the Hamming distance: the number of bits that are different between two binary vectors

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

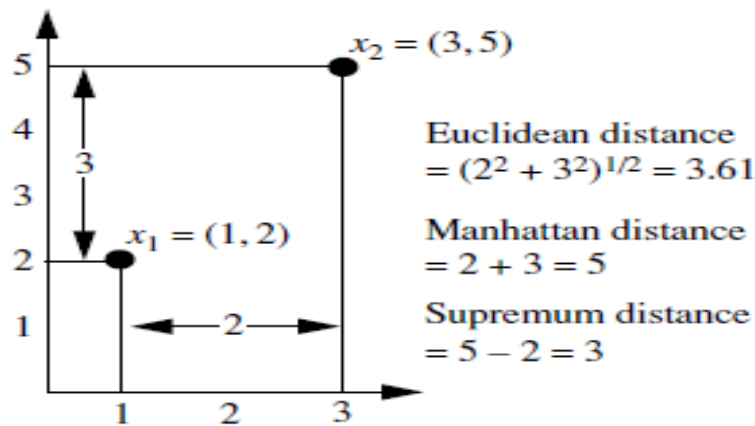
- $h = 2$: (L_2 norm) Euclidean distance

$$d(i, j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \dots + |x_{ip} - x_{jp}|^2)}$$

- $h \rightarrow \infty$. “supremum” ($L_{\max}$ norm, L_{∞} norm) distance.
 - This is the maximum difference between any component (attribute) of the vectors

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f^p |x_{if} - x_{jf}|$$

- The L_1 norm is also known as the *uniform norm*.



Euclidean, Manhattan, and supremum distances between two objects.

Example

Supremum distance. Let's use the same two objects, $x_1 = (1, 2)$ and $x_2 = (3, 5)$, as in Figure above. The second attribute gives the greatest difference between values for the objects, which is $5 - 2 = 3$. This is the supremum distance between both objects.

If each attribute is assigned a weight according to its perceived importance, the **weighted Euclidean distance** can be computed as

$$d(i, j) = \sqrt{w_1|x_{i1} - x_{j1}|^2 + w_2|x_{i2} - x_{j2}|^2 + \dots + w_m|x_{ip} - x_{jp}|^2}.$$

→ Weighting can also be applied to other distance measures as well.

Cosine Similarity

- A **document** can be represented by thousands of attributes, each recording the *frequency* of a particular word (such as keywords) or phrase in the document.

Document	team	coach	hockey	baseball	soccer	penalty	score	win	loss	season
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Other vector objects: gene features in micro-arrays, ...
- Applications: information retrieval, biologic taxonomy, gene feature mapping, ...
- Cosine measure: If d_1 and d_2 are two vectors (e.g., term-frequency vectors), then

$$\cos(d_1, d_2) = (d_1 \bullet d_2) / \|d_1\| \|d_2\| ,$$

Example: Cosine Similarity

- $\cos(d_1, d_2) = (d_1 \bullet d_2) / \|d_1\| \|d_2\|$,
where $\bullet$ indicates vector dot product, $\|d\|$: the length of vector d
- Ex: Find the **similarity** between documents 1 and 2.

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$d_1 \bullet d_2 = 5*3+0*0+3*2+0*0+2*1+0*1+0*1+2*1+0*0+0*1 = 25$$

$$\|d_1\| = \sqrt{(5*5+0*0+3*3+0*0+2*2+0*0+0*0+2*2+0*0+0*0)}^{0.5} = \sqrt{42}^{0.5} = 6.481$$

$$\|d_2\| = \sqrt{(3*3+0*0+2*2+0*0+1*1+1*1+0*0+1*1+0*0+1*1)}^{0.5} = \sqrt{17}^{0.5}$$

62

DATA VISUALIZATION

Why data visualization?

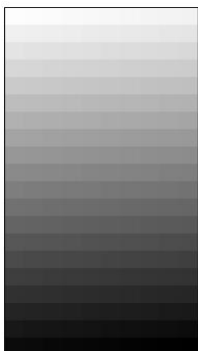
- Gain insight into an information space by mapping data onto graphical primitives
- Provide qualitative overview of large data sets
- Search for patterns, trends, structure, irregularities, relationships among data
- Help find interesting regions and suitable parameters for further quantitative analysis
- Provide a visual proof of computer representations derived

Categorization of visualization methods:

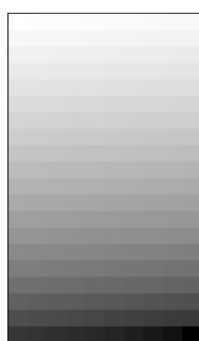
1. Pixel-oriented visualization techniques
2. Geometric projection visualization techniques
3. Icon-based visualization techniques
4. Hierarchical visualization techniques
5. Visualizing complex data and relations

1. Pixel-oriented visualization techniques

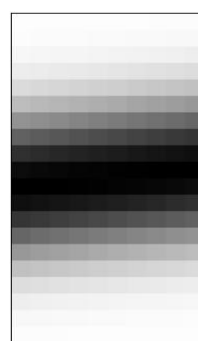
- For a data set of m dimensions, create m windows on the screen, one for each dimension
- The m dimension values of a record are mapped to m pixels at the corresponding positions in the windows
- The colors of the pixels reflect the corresponding values



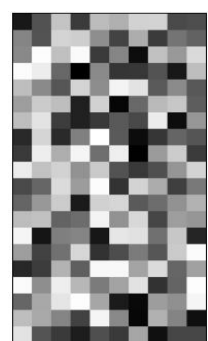
(a) Income



(b) Credit Limit



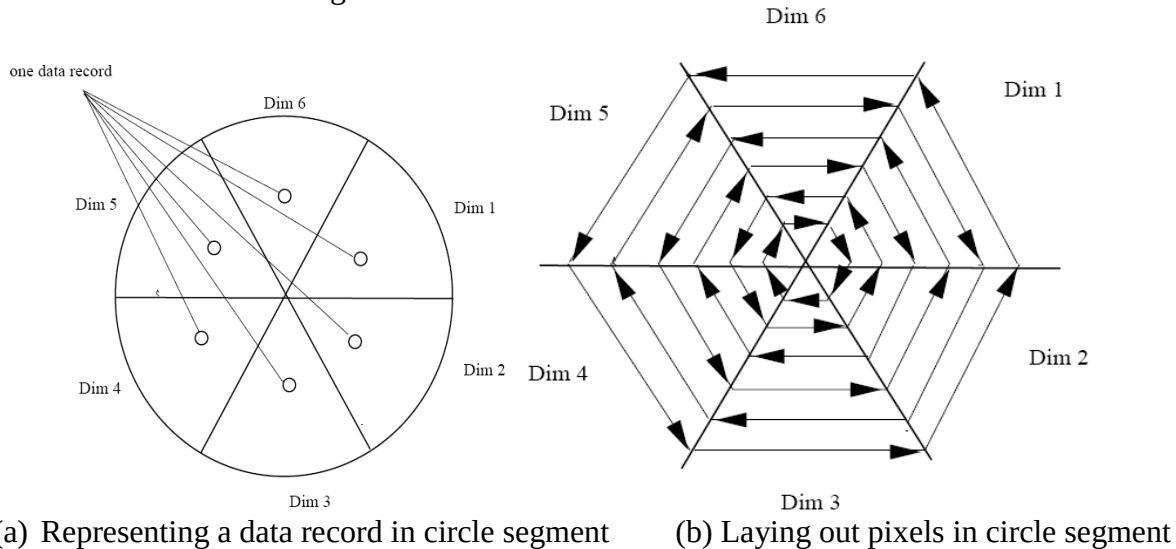
(c) transaction volume



(d) age

Laying out pixels in circle segments

→ To save space and show the connections among multiple dimensions, space filling is often done in a circle segment

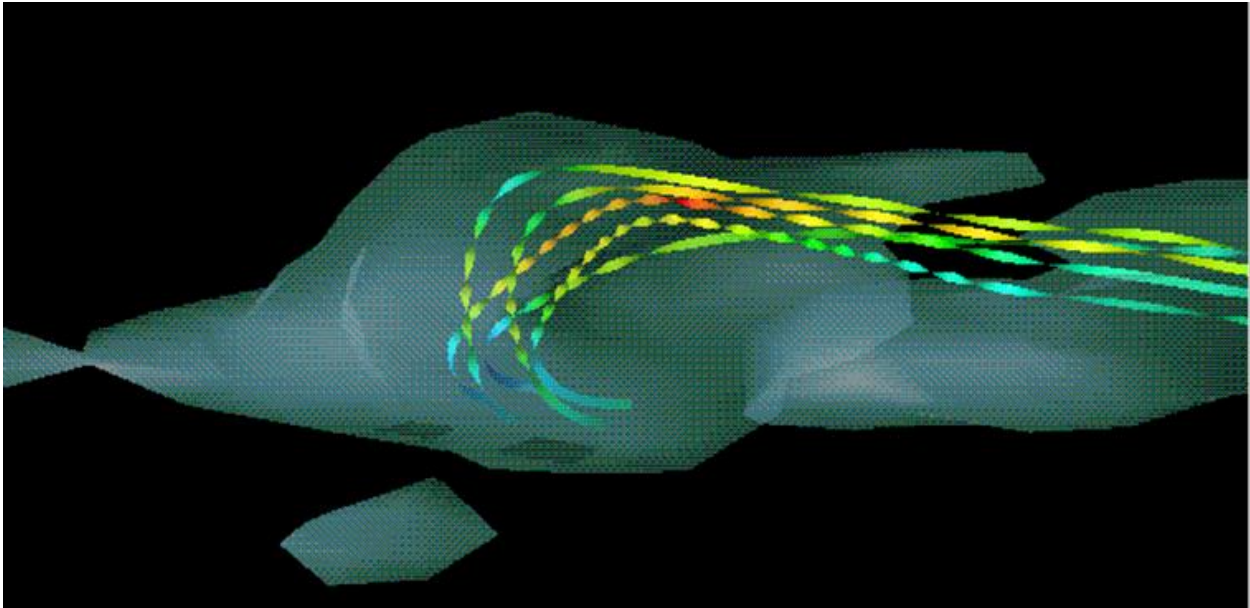


2. Geometric projection visualization techniques- → Visualization of geometric transformations and projections of the data

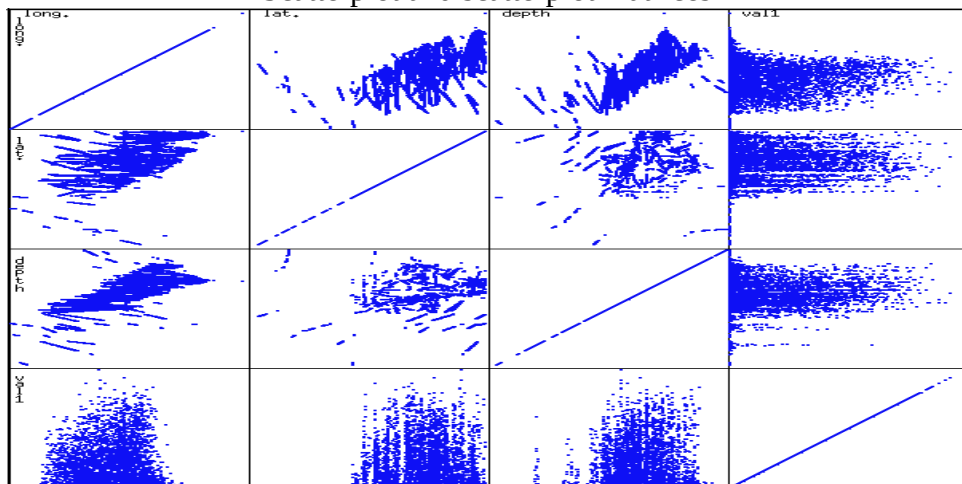
Different Methods are

- Direct visualization
- Scatterplot and scatterplot matrices
- Landscapes
- Projection pursuit technique: Help users find meaningful projections of multidimensional data
- Prosecution views
- Hyperslice
- Parallel coordinates

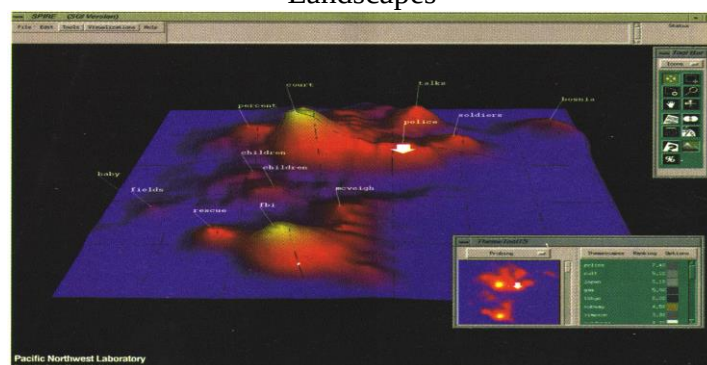
Direct visualization



Scatterplot and scatterplot matrices



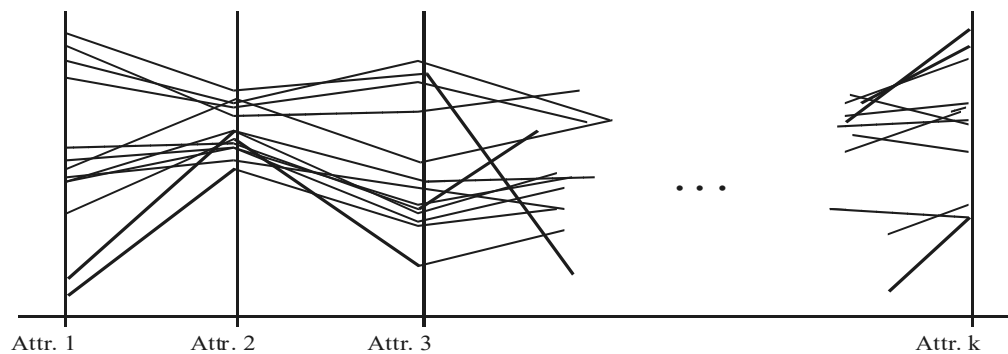
Landscapes



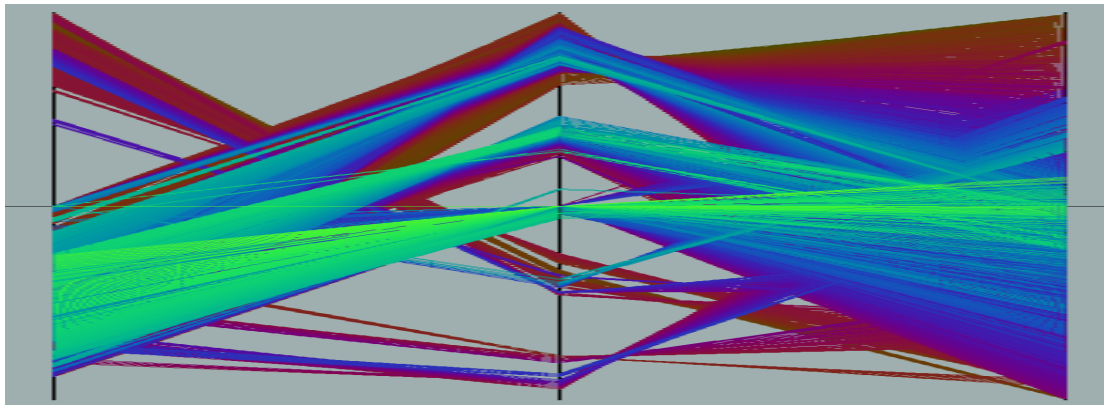
- Visualization of the data as perspective landscape
- The data needs to be transformed into a (possibly artificial) 2D spatial representation which preserves the characteristics of the Projection pursuit technique: Help users find meaningful projections of multidimensional data

Parallel coordinates

- n equidistant axes which are parallel to one of the screen axes and correspond to the attributes
- The axes are scaled to the [minimum, maximum]: range of the corresponding attribute
- Every data item corresponds to a polygonal line which intersects each of the axes at the point which corresponds to the value for the attribute



Parallel Coordinates of a Data Set



3. Icon-based visualization techniques

- Visualization of the data values as features of icons
- Typical visualization methods
 - Chernoff Faces
 - Stick Figures
- General techniques
 - Shape coding: Use shape to represent certain information encoding
 - Color icons: Use color icons to encode more information
 - Tile bars: Use small icons to represent the relevant feature vectors in document retrieval

Chernoff Faces Stick Figures

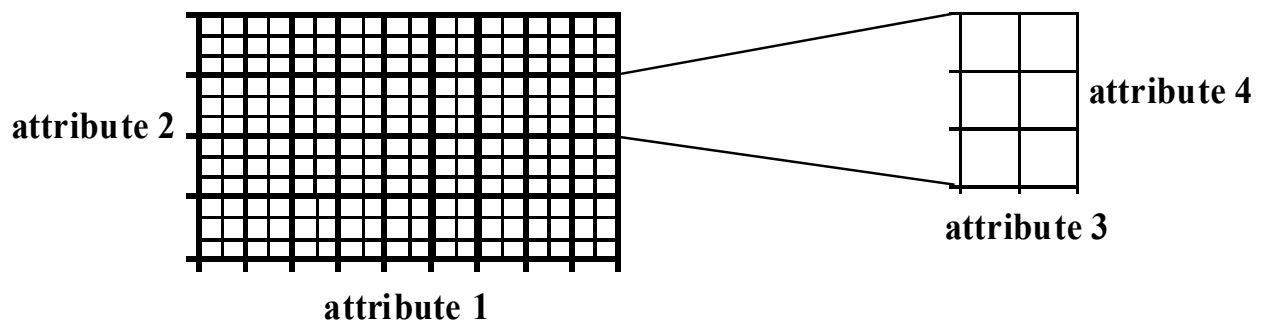


4. Hierarchical visualization techniques

- Visualization of the data using a hierarchical partitioning into subspaces
- Methods

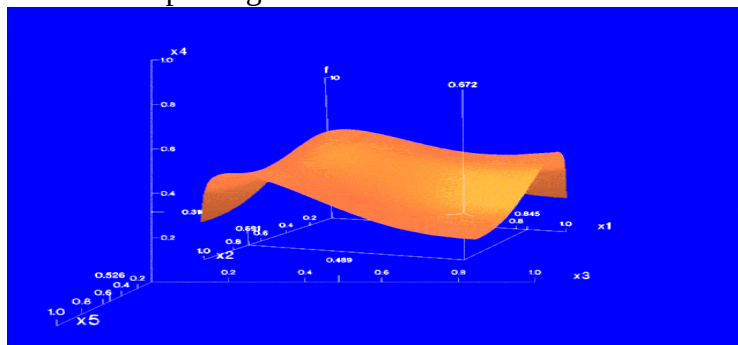
- **Dimensional Stacking**

- Partitioning of the n-dimensional attribute space in 2-D subspaces, which are 'stacked' into each other
- Partitioning of the attribute value ranges into classes. The important attributes should be used on the outer levels.
- Adequate for data with ordinal attributes of low cardinality
- But, difficult to display more than nine dimensions
- Important to *map dimensions appropriately*



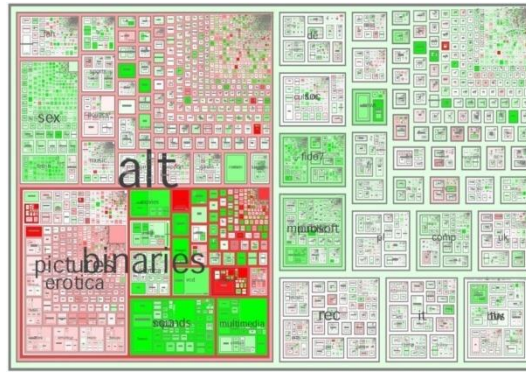
- **Worlds-within-Worlds**

- Assign the function and two most important parameters to innermost world
- Fix all other parameters at constant values - draw other (1 or 2 or 3 dimensional worlds choosing these as the axes)
- Software that uses this paradigm

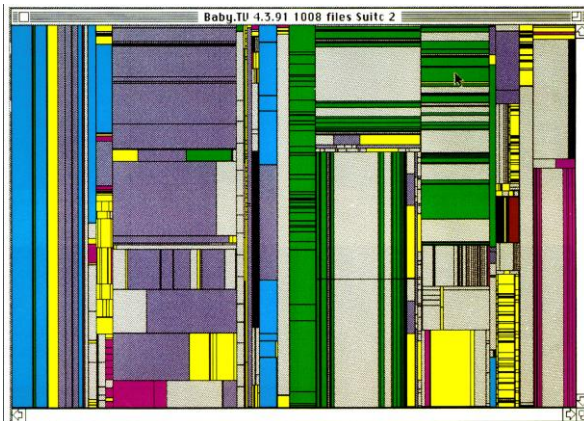


- **Tree-Map**

- Screen-filling method which uses a hierarchical partitioning of the screen into regions depending on the attribute values
- The x- and y-dimension of the screen are partitioned alternately according to the attribute values (classes)



▪ **Cone Trees**



InfoCube

- A 3-D visualization technique where hierarchical information is displayed as nested semi-transparent cubes
- The outermost cubes correspond to the top level data, while the subnodes or the lower level data are represented as smaller cubes inside the outermost cubes, and so on



5. Visualizing complex data and relations

- Visualizing non-numerical data: text and social networks
- Tag cloud: visualizing user-generated tags
 - The importance of tag is represented by font size/color
- Besides text data, there are also methods to visualize relationships, such as visualizing social networks

Consequently, the N observations that we have of X are represented as follows: $x_1, x_2, \dots, x_N$. Within the scope of this discussion, the compilation of numerical information is typically referred to as the "data set" (for X). On what part of the scatter plot would the majority of the salary data points be located? Because of this, we are able to gain a sense of the general pattern of the data.

Indicators of central tendency include the mean, median, mode, and even the data's midpoint.

The (arithmetic) mean is the most common and trustworthy numerical measure of the "center" of a data set. It is also one of the most often used measures of central tendency. Let's assume that X is a numeric property and that $x_1, x_2, \dots, x_N$ are the N values or observations. The sum total of these numbers is equal to

$$\text{MEAN} = \sum N \div n$$

where N is the sum of all integers

n represents the total number of integers.

DATA: { 16, 17, 10, 13, 20, 18, 13, 14, 18 }

$$\bar{x} = \frac{16 + 17 + 10 + 13 + 20 + 18 + 13 + 14 + 18}{9}$$

$$\bar{x} = \frac{139}{9}$$

$$\bar{x} = 15.444 \text{ (rounded to three decimal places)}$$

Median

While there is no more descriptive statistic than the mean, it is not necessarily the most accurate approach to locating the data center. The mean's extreme value (or outlier) sensitivity is a serious flaw. Even a handful of outliers can skew the median. For instance, some companies may have one or two exceptionally well-paid managers who drive up the average income for everyone else. The same holds true for test results: a small number of students with extremely poor marks can significantly lower the class average. The trimmed mean, which is the mean found by cutting off the values at the high and low ends, can be used to reduce the effect of a tiny number of extreme values. In order to find the average income, for instance, we may filter the data and discard the highest and lowest numbers. Avoiding a 20% snip at each end might save us from losing some crucial details.

The [median](#), the middle value in a set of ordered data values, is a more appropriate measure of the center of data when the data are skewed (asymmetric). It's the threshold that divides the top half from the bottom half of a dataset.

While the median is typically used with numerical data in probability and statistics, applying the notion to ordinal data is also possible. For the sake of argument, let's say that N values for attribute X have been sorted ascendingly in a given data collection. The median of an ordered collection is the middle value if N is odd. In the case where N is an even number, the median is not a discrete value but rather the midpoint between the two extremes.

The median is calculated as the mean of the middle two values of a numerical attribute X.

Mode

The mode, in addition to the mean, can be utilized in calculating the average. The mode is the statistic representing the value that occurs the most frequently in a set of numbers.

As a consequence of this, it is possible to compute both its qualitative and quantitative features of it. It is possible for there to be a variety of modes since the frequency with the highest value might have a number of different values. Data sets can be classified as unimodal, bimodal, or trimodal, depending on the number of modes they include. When talking about multimodal data sets, it is usual practice to refer to them as having two or more modes. On the other hand, if each data value only appears once, there will be no mode.

$$\text{Mean} = \frac{\text{Sum of Observations}}{\text{Total Number of Observations}}$$

If 'n' is odd: $\text{Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term}$

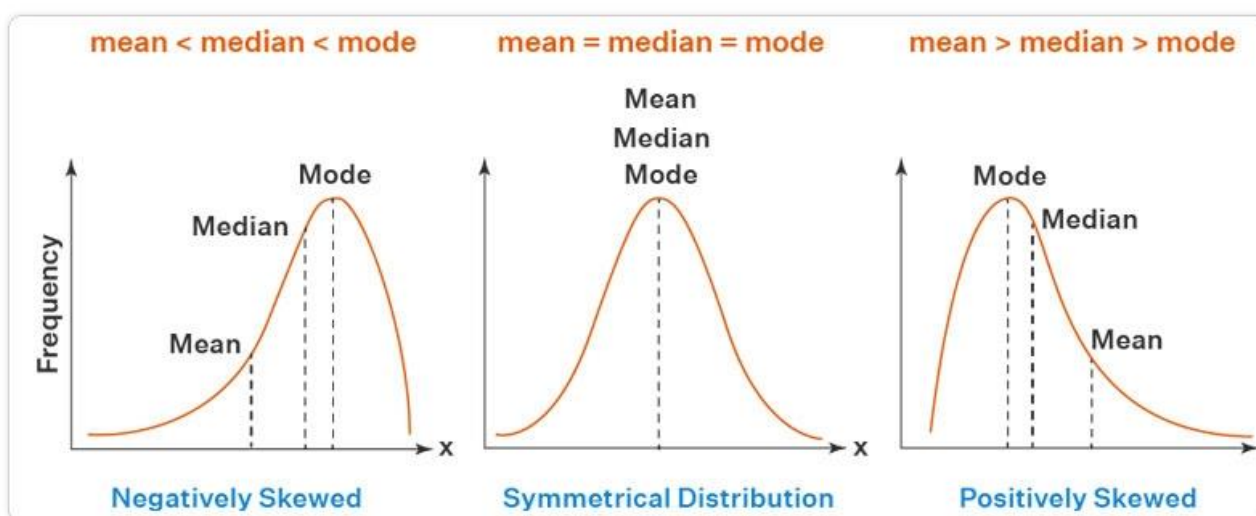
If 'n' is even: $\text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2}$

$$\text{Mode} = L + h \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)}$$

To know why and how to pursue a career in data science, refer to the [data science career path](#).

The following empirical connection holds for unimodal frequency distributions that are somewhat asymmetrical:

$$\text{Mean} - \text{Mode} = 3 \times (\text{Mean} - \text{Median})$$



As a result, when the mean and median are available, calculating the mode of a unimodal frequency distribution that is only slightly skewed is a piece of cake.

As shown in the Figure above, the mean, median, and mode are all centered on the same value in a unimodal frequency curve exhibiting perfect symmetry in the data. Unfortunately, though, the facts in most practical contexts are asymmetric. It's also possible for them to be positively skewed, with the mode below the median, or negatively skewed, with the mode above the median.

The central tendency of data collection may also be evaluated using the midrange. It is calculated by averaging the greatest and lowest numbers in the set. SQL's `max()` and `min()` aggregate methods make it simple to calculate this algebraic metric ().

A lot of people are confused about the role of a Data Scientist and a Data Analyst; even though both deal with “Data,” there are still a good number of significant differences between them. Do you want to know the clear [difference between a data scientist and a data analyst](#), click here.

The Dispersion of Data

Before diving into further measures of data dispersion, let's first examine the range, quantiles, quartiles, percentiles, and interquartile range.

Range

Let X be a numerical attribute, and let $x_1, x_2, \dots, x_N$ be a series of observations for X . The set's range is calculated by taking the absolute value of the difference between the greatest and smallest values in the set ($\max() - \min()$).

$$\text{Range} = X_{\max} - X_{\min}$$

Quartile

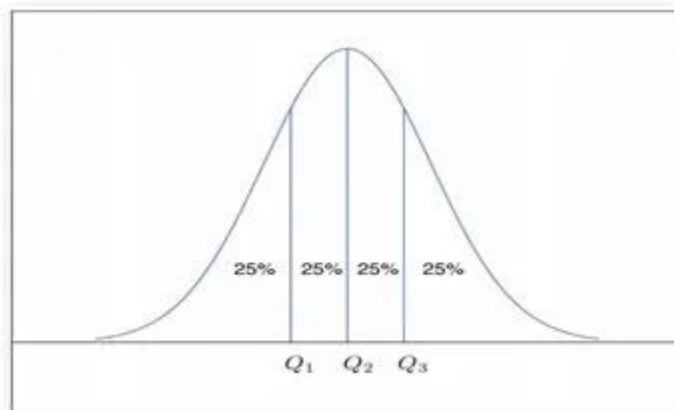
Let's pretend that we have a set of numbers ordered from largest to smallest for attribute X . Imagine we could select particular data points to partition the data distribution equally. Quartiles are a statistical description of data measures based on these values. The quartiles of distribution are discrete numbers chosen at predetermined intervals to create groups of data that are roughly equivalent in size. (We say “*basically*” because there might not be values of X that divide the data into precisely equal-sized sections. For the sake of clarity, let's assume they're on par. For a given distribution of data, the k th q -quartile is the number x such that no more than k/q of the data values are fewer than x , and no more than $(q - k)/q$ of the data values are higher than x , where k is an integer such that $0 \leq k \leq q$. Each of the q -quartiles has a probability of $1/q$).

The 2-quartile is the midpoint between a data set's minimum and maximum values. You may think of it as the middle number. The three data points known as the “4-quartiles” that divide the distribution into four equal parts, with each portion representing one-fourth of the whole, provide the basis for this analysis. Most people will refer to them as “*quartiles*” instead. The median, the quartiles, and the

percentiles are the most often used quartiles, whereas the 100-quartiles are more usually known as percentiles; they divide the data distribution into 100 equal-sized sequential groupings.

The quartiles can be used to determine a distribution's shape, size, and central tendency. Q1 represents the 25th percentile or the first quartile. It discards the bottom 25% of the records.

In this case, Q3 represents the 75th percentile, which eliminates the bottom 75% (or top 25%) of the data. The 50th percentile is the middle quartile. The median represents the midpoint of a set of data.



A straightforward measure of dispersion, the gap between the first and third quartiles reveals the span of values occupied by the middle 50% of the data. The IQR formula is $IQR = Q3 - Q1$, where Q3 and Q1 are the third and first quartiles.

$$\text{Quartile Deviation} = \frac{\text{Third Quartile} - \text{First Quartile}}{2}$$

$$\text{Quartile Deviation (Q.D)} = \frac{Q_3 - Q_1}{2}$$

Variance and Standard Deviation

Distributive metrics of data include variance and standard deviation. They show how dispersed a data set is. When the standard deviation is small, the data points cluster tightly around the mean, but when it's big, the data points are dispersed over a wide range of values.

The [variance](#) of N observations, $x_1, x_2, \dots, x_N$, for a numeric attribute X

The **variance** of a population of n is $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n} = \frac{\sum_{i=1}^n x_i^2}{n} - \mu^2$.

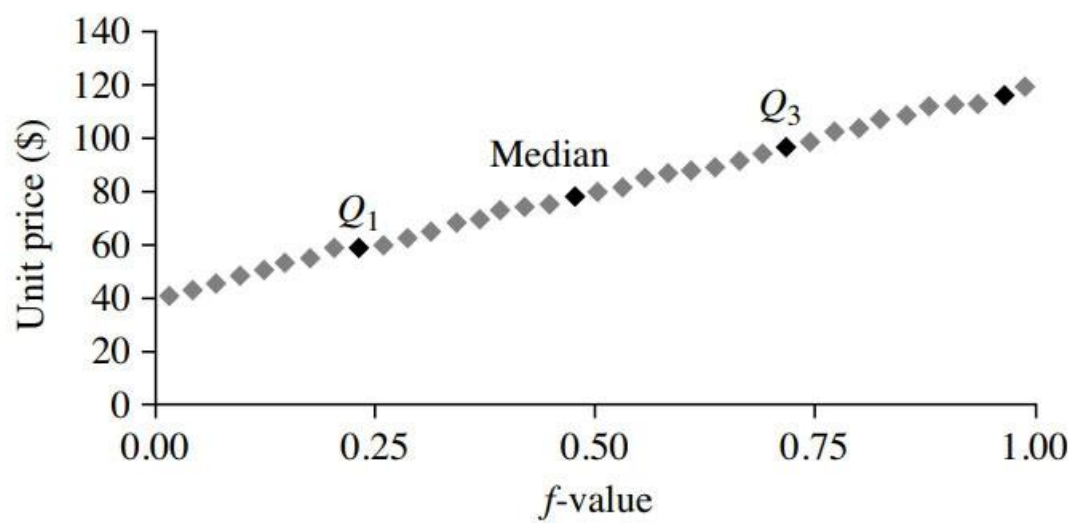
The **standard deviation** of a population of n is $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$.

A Quantile-Quantile Plot

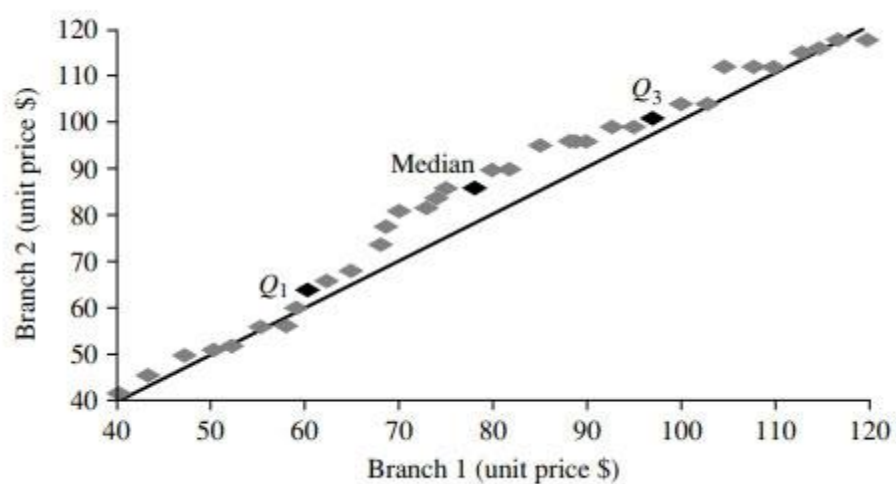
The quantiles of one univariate distribution are shown against the quantiles of another distribution in a q-q graphic. It's an effective mental imagery method since it permits the user to see whether there is a change while switching distributions.

Let's pretend we have data from two branches with respect to the attribute or variable unit price. For simplicity, we will refer to the first branch's data as $x_1, \dots, x_N$ and the second branch's data as $y_1, \dots, y_M$, with both sets of data being arranged from most recent to least recent. In the case where $M = N$ (i.e., the number of points in both sets is equal), we may plot y_i versus x_i , where y_i and x_i are the $(I - 0.5)/N$ quantiles of the two sets of data.

Only M points will fit on the q-q plot if $M < N$ (the second branch contains fewer data than the first). The quantile of the y distribution at index I , expressed as y_i , is given by $(I - 0.5)/M$.



Unit price (\$)	Count of items sold
40	275
43	300
47	250
—	—
74	360
75	515
78	540
—	—
115	320
117	270
120	350



Distribution of the quantiles vs the quantiles. Plots unit pricing data on a quantile-quantile retail sales chart at two AllElectronics locations during a specific time frame. For each data set, each dot represents a quantile, and the unit prices at branch 1 are shown against those at branch 2 at that quantile. (The straight line is meant to be compared to when the unit price is the same at all branches for a particular quantile. The darker data points represent information from Quarter 1, the median, and Quarter 3. For example, we can observe that in the first quarter of this year, the unit price of things sold at branch 1 was somewhat lower than that at branch 2. So, 25% of sales at Store 1 were for products that cost less than equivalent to \$60, but at Branch 2, only 25% of things sold were \$60 or less. Half of the things sold at Branch 1 were less than \$78 (the median, see Q2), and half sold at Branch 2 were less than \$85 (the 25th percentile). Generally, we see that the unit prices of goods sold at branch 1 are lower than those sold at branch 2, indicating a change in distribution.

Histograms

The histogram, sometimes known as a frequency histogram, has been used for at least a century. The word "*histogram*" comes from the Greek words for "*pole*" and "*gramme*," therefore, a histogram is literally a chart of poles. Histograms are a graphical representation for summing up the distribution of an individual value of an attribute X. For each known value, a pole or vertical bar is drawn if X is nominal, like a car model or item category. Each bar's height represents the number of times that particular X value occurred. This type of chart is most generally referred to as a bar chart.

Histogram is the recommended phrase if X is a numerical variable. Separate, sequential subranges of X values have been established. The subranges represent separate segments of X's data distribution, sometimes known as buckets or bins. A bucket's breadth refers to its reach. In most cases, each bucket has the same width as the others. A price characteristic with values from \$1 (rounded up to the closest dollar) to \$200 (also rounded up to the nearest dollar) can be divided into the subranges 1-20, 21-40, 42-60, etc.

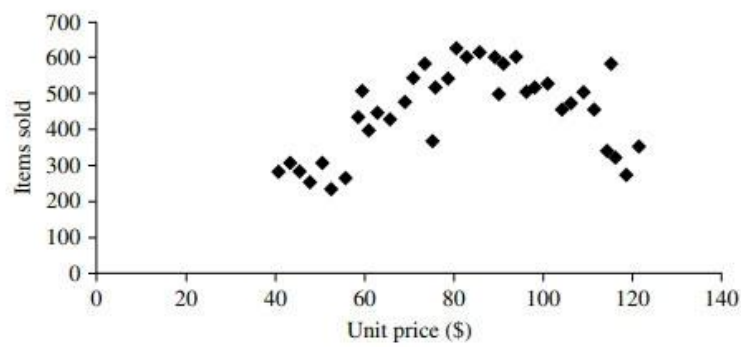
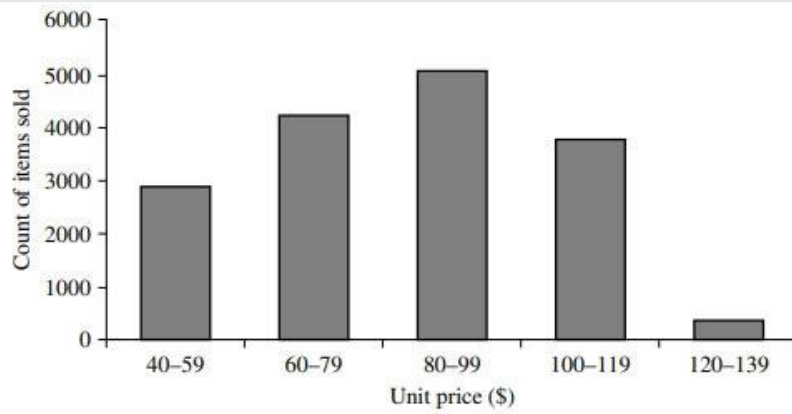
Each subrange is represented by a bar whose height is proportional to the number of data points that fall within that subrange.

Data Correlation and Scatter Plots

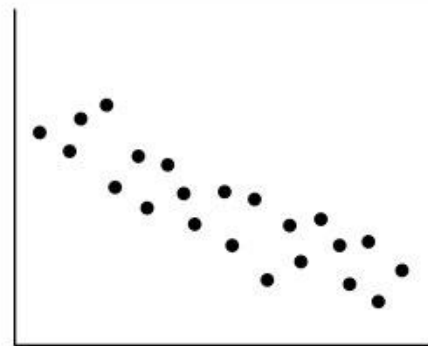
In order to determine if there is a correlation between two variables, a scatter plot is a useful graphical tool. Apparent connection, pattern, or development between the two quantitative features. Each pair of values is interpreted as an algebraic pair of coordinates and represented as a point on the graph paper. The data in Figure are plotted as a scatter diagram. Table

The scatter plot is an effective tool for quickly seeing patterns in bivariate data, identifying outliers, and investigating possible correlations.

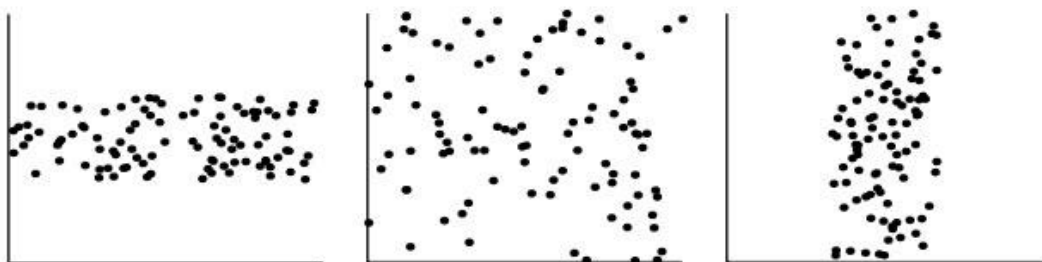
When one characteristic (X) implies the other (Y), we say the two characteristics are correlated. A correlation can be either positive or negative or even nonexistent (uncorrelated). Examples of positive and negative relationships between the two characteristics are displayed in the figure. When a pattern of points is plotted, a slope is indicated if the points. Since X is increasing while Y is increasing, a positive correlation may be inferred from the data (Figure). Given a plotted pattern,



(a)



(b)



Slopes from left to right, indicating a negative association between X and Y; X values rise as Y values fall (Figure). A line of best fit can be drawn to investigate the link between the variables. You may find information about correlation tests using statistics. In the data sets depicted in each figure, there are three instances in which a correlation exists between the two qualities.

Measuring the Central Tendency: Mean, Median, and Mode

The most common and effective numeric measure of the “center” of a set of data is the (*arithmetic*) *mean*. Let $x_1, x_2, \dots, x_N$ be a set of N values or *observations*, such as for some numeric attribute X , like *salary*. The **mean** of this set of values is

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}.$$

Example 2.6 Mean. Suppose we have the following values for *salary* (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110. Using Eq. (2.1), we have

$$\begin{aligned}\bar{x} &= \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12} \\ &= \frac{696}{12} = 58.\end{aligned}$$

Thus, the mean salary is \$58,000. ■

Sometimes, each value x_i in a set may be associated with a weight w_i for $i = 1, \dots, N$. The weights reflect the significance, importance, or occurrence frequency attached to their respective values. In this case, we can compute

$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_N x_N}{w_1 + w_2 + \dots + w_N}.$$

This is called the **weighted arithmetic mean** or the **weighted average**.

Statistical descriptions help summarize and understand the main features of a data set.

1. Measures of Central Tendency- Mean, Median, and Mode

These describe the **center** or typical value of a dataset.

a. Mean (Average)

Mean =
$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N}.$$

Example:

Data: 4, 8, 6, 5, 3

Mean = $(4 + 8 + 6 + 5 + 3) / 5 = 26 / 5 = 5.2$

b. Median

The **middle value** when data is sorted.

Example:

Data: 3, 4, 5, 6, 8 (already sorted)

Median = 5

c. Mode

The value that occurs **most frequently**.

Example:

Data: 4, 5, 6, 5, 3

Mode = 5 (appears twice)

2. Measures of Dispersion (Spread)

They describe how **spread out** the data is.

a. Range

Range = Max - Min $\text{Range} = \text{Max} - \text{Min}$

Example:

Data: 3, 4, 5, 6, 8

Range = $8 - 3 = 5$

b. Variance (σ^2)

Average of squared differences from the Mean.

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Example:

Data: 2, 4, 6

Mean = 4

$$\text{Variance} = [(2-4)^2 + (4-4)^2 + (6-4)^2] / 3 = (4 + 0 + 4) / 3 = 2.67$$

c. Standard Deviation (σ)

The **square root** of the variance.

$$\sigma = \sqrt{\sigma^2}$$

From above example:

$$\text{Standard Deviation} = \sqrt{2.67} \approx 1.63$$

3. Skewness

Describes **asymmetry** in the distribution.

- **Positive skew:** Tail on the right (mean > median)
- **Negative skew:** Tail on the left (mean < median)
- **Symmetric:** Mean $\approx$ Median

4. Kurtosis

Describes **tailedness** of the distribution.

- **High kurtosis:** More outliers (peaked)
- **Low kurtosis:** Flatter distribution

5. Five-Number Summary

1. Minimum
2. First Quartile (Q1)

3. Median (Q2)
4. Third Quartile (Q3)
5. Maximum

Useful for creating **boxplots**.

Example Dataset:

Data: 2, 4, 4, 5, 7, 8, 9

- Mean = $(2+4+4+5+7+8+9)/7 = 5.57$
- Median = 5
- Mode = 4
- Range = $9 - 2 = 7$
- Q1 = 4
- Q3 = 8